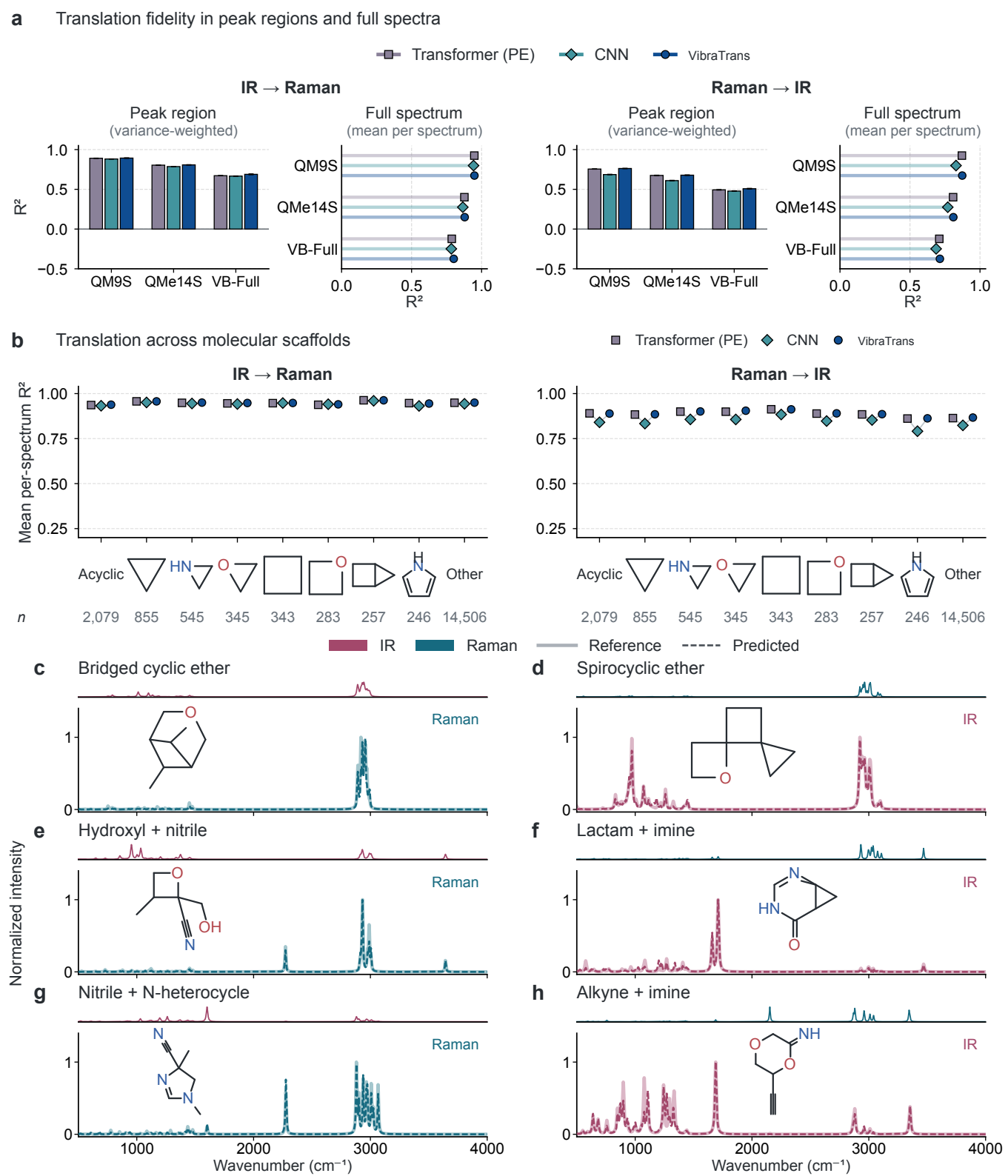

a
Translation fidelity in peak regions and full spectra
Transformer (PE)
CNN
VibraTrans
IR → Raman
Raman → IR
Peak region
(variance-weighted)
Full spectrum
(mean per spectrum)
R²
QM9S
QMe14S
VB-Full
b
Translation across molecular scaffolds
Mean per-spectrum R²
Acyclic
Other
n
2,079
855
545
345
343
283
257
246
14,506
IR
Raman
Reference
Predicted
c
Bridged cyclic ether
d
Spirocyclic ether
e
Hydroxyl + nitrile
f
Lactam + imine
g
Nitrile + N-heterocycle
h
Alkyne + imine
Normalized intensity
Wavenumber (cm⁻¹)